

3.9 Lagrange Multipliers Worksheet

1. Suppose ACME's revenue (in thousands of dollars) from the sale of x widgets and y gadgets can be expressed by $R(x, y) = 150x - x^2 + 120y - \frac{1}{2}y^2$. Suppose that creating a widget requires 4 gizmos and gadget requires 2 gizmos. ACME only has 120 gizmos available and wants to use all of them. Determine ACME's maximum revenue subject to the given constraint.

We know the revenue function $R(x, y) = 150x - x^2 + 120y - \frac{1}{2}y^2$ is subject to the constrain $4x + 2y = 120$, where $g(x, y) = 4x + 2y$.

According to the method of Lagrange Multipliers, we must find x, y , and λ such that:

$$\begin{aligned}\vec{\nabla}R(x, y) &= \lambda \vec{\nabla}g(x, y) \text{ and } 4x + 2y = 120 \\ \Rightarrow \langle 150 - 2x, 120 - y \rangle &= \lambda \langle 4, 2 \rangle \text{ and } 4x + 2y = 120.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 150 - 2x = 4\lambda \\ 120 - y = 2\lambda \\ 4x + 2y = 120 \end{cases}$$

From the first two equations we can see that:

$$\begin{cases} \frac{150 - 2x}{4} = \lambda \\ \frac{120 - y}{2} = \lambda \end{cases} \rightarrow \frac{150 - 2x}{4} = \frac{120 - y}{2} \rightarrow 300 - 4x = 480 - 4y \rightarrow -4x + 4y = 180$$

If we combine this with the last equation from the original system, we obtain:

$$\begin{cases} 4x + 2y = 120 \\ -4x + 4y = 180 \end{cases} \rightarrow 6y = 300 \rightarrow y = 50 \rightarrow x = 5$$

If $y = 50$ & $x = 5$ this implies that $\lambda = 35$. Therefore $R(5, 50) = 5475$ is either an absolute minimum or maximum. However, we see that trivially $(30, 0)$ also satisfies the constraint and would produce a revenue of $R(30, 0) = 3600$ so we can conclude that $R(5, 50) = 5475$ (*thousands of dollars*) = \$5,475,000 is the absolute maximum revenue given the constraint.

2. Find the points on the surface $xy - z^2 = 1$ that are closest to the origin.

We are attempting to minimize distance $d = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2}$, but remember that $\sqrt{f(x)}$ is minimized when $f(x)$ is minimized, so it suffices to minimize where $D(x, y, z) = x^2 + y^2 + z^2$.

We are working to minimize distance subject to the constraint $xy - z^2 = 1$ where $g(x, y, z) = xy - z^2$.

According to the method of Lagrange Multipliers, we must find x, y, z , and λ such that:

$$\begin{aligned}\bar{\nabla} D(x, y, z) &= \lambda \bar{\nabla} g(x, y, z) \text{ and } xy - z^2 = 1 \\ \Rightarrow \langle 2x, 2y, 2z \rangle &= \lambda \langle y, x, -2z \rangle \text{ and } xy - z^2 = 1.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2x = \lambda y \\ 2y = \lambda x \\ 2z = -2\lambda z \\ xy - z^2 = 1 \end{cases}$$

From the third equation we can deduce that $\lambda = -1$ or $z = 0$.

If $\lambda = -1$, we know from the first two equations that $\begin{cases} 2x = -y \\ 2y = -x \end{cases} \rightarrow 2(-2x) = -x \rightarrow -3x = 0 \rightarrow x = 0$

and therefore $y = 0$. However, if this is the case then by the last equation we must have $-z^2 = 1 \rightarrow z^2 = -1$ which is clearly impossible! Therefore, we know that $\lambda \neq -1$ and so it must be the case that $z = 0$.

If $z = 0$ then we can reduce the original system to $\begin{cases} 2x = \lambda y \\ 2y = \lambda x \\ xy = 1 \end{cases}$.

From here we can rewrite the first two equations to solve for λ .

$$\begin{cases} \frac{2x}{y} = \lambda \\ \frac{2y}{x} = \lambda \end{cases} \rightarrow \frac{2x}{y} = \frac{2y}{x} \rightarrow 2x^2 = 2y^2 \rightarrow x = \pm y$$

If $x = -y$ then this violates the equation $xy = 1$. So we must have $x = y$

So now we have that $\begin{cases} x = y \\ xy = 1 \end{cases}$. From here we can deduce that $x = \pm 1$ & $y = \pm 1$.

Therefore, $D(1, 1, 0) = 2$ & $D(-1, -1, 0) = 2$ may be a potential maximum/minimum of the function.

However, we see that trivially $\left(2, \frac{1}{2}, 0\right)$ also satisfies the constraint and would produce a D value of

$D\left(2, \frac{1}{2}, 0\right) = 4 + \frac{1}{4} = 4.25$, so we can conclude that 2 is the absolute minimum of value of D and therefore $\sqrt{2}$ is the minimum distance for points on the surface to the origin.

3. Let $T(x, y, z) = 20 + 2x + 2y + z^2$ represent the temperature at each point on the sphere $x^2 + y^2 + z^2 = 11$. Use Lagrange multipliers to find the extreme temperatures on this surface.

We know the temperature function $T(x, y, z) = 20 + 2x + 2y + z^2$ is subject to the constrain $x^2 + y^2 + z^2 = 11$, where $g(x, y, z) = x^2 + y^2 + z^2$.

According to the method of Lagrange Multipliers, we must find x, y, z , and λ such that:

$$\begin{aligned}\vec{\nabla}T(x, y, z) &= \lambda \vec{\nabla}g(x, y, z) \text{ and } x^2 + y^2 + z^2 = 11 \\ \Rightarrow \langle 2, 2, 2z \rangle &= \lambda \langle 2x, 2y, 2z \rangle \text{ and } x^2 + y^2 + z^2 = 11.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2 = 2x\lambda \\ 2 = 2y\lambda \\ 2z = 2z\lambda \\ x^2 + y^2 + z^2 = 11 \end{cases}$$

We see from the third equation that $z = 0$ and/or $\lambda = 1$

If $\lambda = 1$ then the first and second equation give us that $x = 1$ & $y = 1$. Then by the third equation we must have that $z^2 = 9 \rightarrow z = \pm 3$.

If $z = 0$ then by what we showed above $\lambda \neq 1$. Then by the first two equations we have that $\lambda = \frac{1}{x}$ & $\lambda = \frac{1}{y}$

which implies that $\frac{1}{x} = \frac{1}{y} \rightarrow y = x$. By the final equation we then have that $2x^2 = 11 \rightarrow x = \pm\sqrt{\frac{11}{2}}$.

Therefore, $(1, 1, \pm 3)$ and $\left(\pm\sqrt{\frac{11}{2}}, \pm\sqrt{\frac{11}{2}}, 0\right)$ are all possible candidates for the absolute maximum and minimum points.

We see that $T(1, 1, \pm 3) = 33$

$$T\left(\sqrt{\frac{11}{2}}, \sqrt{\frac{11}{2}}, 0\right) = 20 + 4\sqrt{\frac{11}{2}}$$

$$T\left(-\sqrt{\frac{11}{2}}, -\sqrt{\frac{11}{2}}, 0\right) = 20 - 4\sqrt{\frac{11}{2}} \text{ and}$$

So we conclude that $T(1, 1, \pm 3) = 33$ is the absolute maximum temperature on the surface and

$T\left(\sqrt{\frac{11}{2}}, \sqrt{\frac{11}{2}}, 0\right) = 20 + 4\sqrt{\frac{11}{2}}$ $T\left(\sqrt{\frac{11}{2}}, -\sqrt{\frac{11}{2}}, 0\right) = T\left(-\sqrt{\frac{11}{2}}, +\sqrt{\frac{11}{2}}, 0\right) = 20$ is the absolute minimum temperature on the surface.

4. Use Lagrange multipliers to find the maximum and minimum values of the function $f(x, y) = x^2 + y^2 + z^2$ subject to the constraint $x^2 + y^2 + 2z = 8$.

We note that $g(x, y, z) = x^2 + y^2 + 2z$. According to the method of Lagrange Multipliers, we must find x, y, z , and λ such that:

$$\begin{aligned}\vec{\nabla} f(x, y, z) &= \lambda \vec{\nabla} g(x, y, z) \text{ and } x^2 + y^2 + 2z = 8 \\ \Rightarrow \langle 2x, 2y, 2z \rangle &= \lambda \langle 2x, 2y, 2 \rangle \text{ and } x^2 + y^2 + 2z = 8.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2x = 2x\lambda \\ 2y = 2y\lambda \\ 2z = 2\lambda \\ x^2 + y^2 + 2z = 8 \end{cases}$$

From the third equation we can deduce that $z = \lambda$.

From the first and second equations we can see that $\lambda = 1$ or $x = 0$ & $y = 0$.

If $x = 0$ & $y = 0$ then by the final equation we must have $2z = 8 \rightarrow z = 4$.

However if $\lambda = 1$ then $z = 1$ and $x^2 + y^2 = 6$.

Therefore, $(0, 0, 4)$ and $(x, y, 1)$ where $x^2 + y^2 = 6$ are all possible candidates for the absolute maximum and minimum points.

We see that $f(0, 0, 4) = 16$ & $f(x, y, 1) = 7$ and so we conclude that $f(0, 0, 4) = 16$ is the absolute maximum of the f and $f(x, y, 1) = 7$ is the absolute minimum of f subject to the constraint.

5. Find the points on the cone $z^2 = x^2 + y^2$ that are closest to the point $(4, 2, 0)$.

We are attempting to minimize distance $d = \sqrt{(x-4)^2 + (y-2)^2 + (z-0)^2}$, but remember that $\sqrt{f(x)}$ is minimized when $f(x)$ is minimized, so it suffices to minimize where $D(x, y, z) = (x-4)^2 + (y-2)^2 + z^2$. We are working to minimize distance subject to the constraint $z^2 = x^2 + y^2$ where $g(x, y, z) = z^2 - x^2 - y^2$.

According to the method of Lagrange Multipliers, we must find x, y, z , and λ such that:

$$\begin{aligned}\bar{\nabla} D(x, y, z) &= \lambda \bar{\nabla} g(x, y, z) \text{ and } z^2 = x^2 + y^2 \\ \Rightarrow \langle 2(x-4), 2(y-2), 2z \rangle &= \lambda \langle -2x, -2y, 2z \rangle \text{ and } z^2 = x^2 + y^2.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2x - 8 = -2x\lambda \\ 2y - 4 = -2y\lambda \\ 2z = 2z\lambda \\ z^2 = x^2 + y^2 \end{cases}$$

From the third equation we can deduce that $z = 0$ or $\lambda = 1$.

Notice that if $z = 0$ then by the last equation $0 = x^2 + y^2 \rightarrow x = y = 0$ and this makes it impossible for the first two equations to be satisfied. Therefore, we can conclude that $z \neq 0$.

So it must be that $\lambda = 1$. Then from the first two equations we have:

$$\begin{cases} 2x - 8 = -2x \\ 2y - 4 = -2y \end{cases} \rightarrow \begin{cases} 4x = 8 \\ 4y = 4 \end{cases} \rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$$

Which then implies that $z^2 = x^2 + y^2 \rightarrow z^2 = 4 + 1 \rightarrow z = \pm\sqrt{5}$

Therefore, $(2, 1, \pm\sqrt{5})$ are possible candidates for the closest points to on the cone to $(4, 2, 0)$.

We see that $D(2, 1, -\sqrt{5}) = 10$ & $D(2, -1, \sqrt{5}) = 10$ may be a potential maximum/minimum of the function.

However, we see that trivially $(0, 0, 0)$ also satisfies the constraint and would produce a D value of

$D(0, 0, 0) = 20$, so we can conclude that 10 is the absolute minimum of value of D and therefore $\sqrt{10}$ is the minimum distance for points on the surface to the origin.

6. Use Lagrange multipliers to find the maximum and minimum values of the function $f(x, y) = x^2 + y$ subject to the constraint $x^2 + y^2 = 4$.

We note that $g(x, y, z) = x^2 + y^2$. According to the method of Lagrange Multipliers, we must find x , y , and λ such that:

$$\begin{aligned}\vec{\nabla} f(x, y) &= \lambda \vec{\nabla} g(x, y) \text{ and } x^2 + y^2 = 4 \\ \Rightarrow \langle 2x, 1 \rangle &= \lambda \langle 2x, 2y \rangle \text{ and } x^2 + y^2 = 4.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2x = 2x\lambda \\ 1 = 2y\lambda \\ x^2 + y^2 = 4 \end{cases}$$

From the first equation we can see that $\lambda = 1$ or $x = 0$.

If $x = 0$ then $y = \pm 2$ by the final equation.

If $\lambda = 1$ then $y = \frac{1}{2}$ by the second equation and then by the final equation $x^2 = \frac{15}{4} \rightarrow x = \pm \frac{\sqrt{15}}{2}$

Therefore, $(0, \pm 2)$ and $\left(\pm \frac{\sqrt{15}}{2}, \frac{1}{2}\right)$ are all possible candidates for the absolute maximum and minimum points.

We see that $f(0, 2) = 2$, $f(0, -2) = -2$ & $f\left(\pm \frac{\sqrt{15}}{2}, \frac{1}{2}\right) = \frac{17}{4}$ and so we conclude that $f\left(\pm \frac{\sqrt{15}}{2}, \frac{1}{2}\right) = \frac{17}{4}$ is the absolute maximum of the f and $f(0, -2) = -2$ is the absolute minimum of f subject to the constraint.

7. Suppose a company makes x units of item A and y units of item B . Suppose also that their profit is given by the function $P(x, y) = 7x^2y - 520x + 2000y$. If the company must make exactly 39 total units how many units of item A and B should it make to maximize its profit?

We know the profit function $P(x, y) = 7x^2y - 520x + 2000y$ is subject to the constrain $x + y = 39$, where $g(x, y) = x + y$.

According to the method of Lagrange Multipliers, we must find x, y , and λ such that:

$$\begin{aligned}\bar{\nabla}P(x, y) &= \lambda \bar{\nabla}g(x, y) \text{ and } x + y = 39 \\ \Rightarrow \langle 14xy - 520, 7x^2 + 2000 \rangle &= \lambda \langle 1, 1 \rangle \text{ and } x + y = 39.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 14xy - 520 = \lambda \\ 7x^2 + 2000 = \lambda \\ x + y = 39 \end{cases}$$

From the first two equations we can see that: $7x^2 + 2000 = 14xy - 520$

However, from the final equation we know that $y = 39 - x$ and therefore we can state that

$$\begin{aligned}7x^2 + 2000 &= 14xy - 520 \rightarrow 7x^2 + 2000 = 14x(39 - x) - 520 \rightarrow 21x^2 - 546x + 2520 = 0 \\ &\rightarrow x^2 - 26x + 120 = 0 \\ &\rightarrow (x - 6)(x - 20) = 0 \\ &\rightarrow x = 6 \text{ or } x = 20\end{aligned}$$

This implies that $y = 33$ or $y = 19$.

Therefore, $(6, 33)$ and $(20, 19)$ are all possible candidates for the absolute maximum and minimum points.

We see that $P(6, 33) = 71196$ and $P(20, 19) = 80800$ and so we conclude that $P(20, 19) = \$80800$ is the absolute maximum profit subject to the constraint.

8. Suppose you live on the surface $z^2 = x^2 + y^2$. The surface is heated by energy from the sun and the temperature at any given point is given by $T(x, y, z) = x^2 y - z$. Use the Lagrange multiplier method to find the location on the surface with the lowest temperature.

We know the temperature function $T(x, y, z) = x^2 y - z$ is subject to the constrain $z^2 = x^2 + y^2$, where $g(x, y, z) = z^2 - x^2 - y^2$.

According to the method of Lagrange Multipliers, we must find x, y, z , and λ such that:

$$\begin{aligned}\vec{\nabla}T(x, y, z) &= \lambda \vec{\nabla}g(x, y, z) \text{ and } z^2 = x^2 + y^2 \\ \Rightarrow \langle 2xy, x^2, -1 \rangle &= \lambda \langle -2x, -2y, 2z \rangle \text{ and } z^2 = x^2 + y^2.\end{aligned}$$

This means that we must solve the nonlinear system:

$$\begin{cases} 2xy = -2x\lambda \\ x^2 = -2y\lambda \\ -1 = 2z\lambda \\ z^2 = x^2 + y^2 \end{cases}$$

We see that $z \neq 0$ and $\lambda \neq 0$ from the third equation. If $x = 0$ then $-2y\lambda = 0$ by the second equation so therefore, $y = 0$ but then by the final equation it must be that $z = 0$ which contradicts what we stated before and so we conclude that $x \neq 0$. The same line of reasoning happens if we let $y = 0$.

We have established that no value can be zero.

From the third equation we have that $\lambda = \frac{-1}{2z}$ and from the first equation we can say that $y = -\lambda$ which implies that $y = \frac{1}{2z}$.

From the second equation we know that $x^2 = -2y\lambda \rightarrow x^2 = -2\left(\frac{1}{2z}\right)\left(-\frac{1}{2z}\right) \rightarrow x^2 = \frac{1}{2z^2}$.

Since we now have x and y in terms of z we will plug these values into the fourth equation and obtain:

$$z^2 = \frac{1}{2z^2} + \frac{1}{4z^2}$$

Since this is an equation of one variable we will solve for z and then determine x and y .

$$z^2 = \frac{1}{2z^2} + \frac{1}{4z^2} \rightarrow 4z^4 = 2 + 1 \rightarrow z = \pm \sqrt[4]{\frac{3}{4}}$$

Therefore $x^2 = \frac{1}{2\left(\pm \sqrt[4]{\frac{3}{4}}\right)^2} = \frac{1}{\sqrt{3}} \rightarrow x = \pm \frac{1}{\sqrt{3}}$ and $y = \frac{1}{2\left(\pm \sqrt[4]{\frac{3}{4}}\right)} = \pm \frac{\sqrt[4]{4/3}}{2}$. (Notice that we have 8 possible

points here).

At these points $T(x, y, z) = \pm \frac{1}{\sqrt{3}} \left(\frac{\sqrt[4]{4/3}}{2} \right) \mp \sqrt[4]{\frac{3}{4}}$. The minimum of these two values is

$$T(x, y, z) = \frac{1}{\sqrt{3}} \left(\frac{\sqrt[4]{4/3}}{2} \right) - \sqrt[4]{\frac{3}{4}}.$$